Advanced Topics in Cloud Computing

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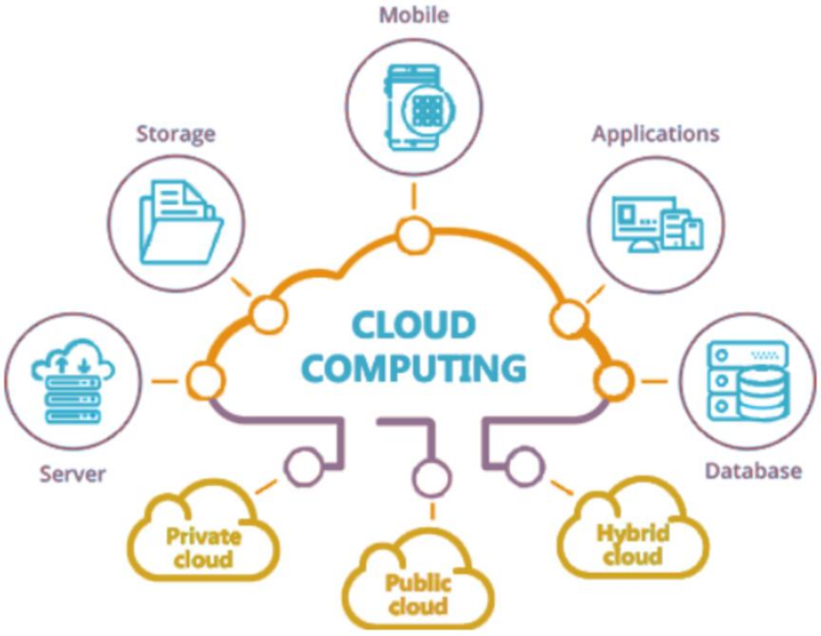
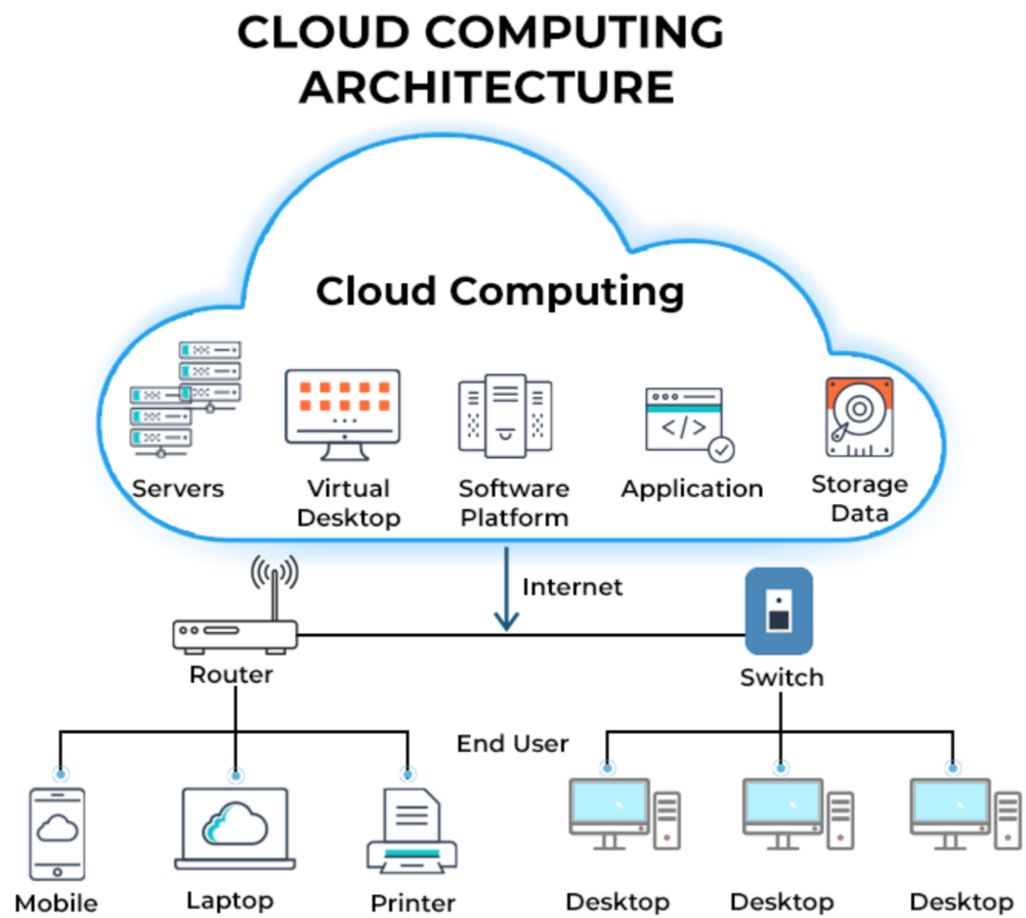
Advanced Topics

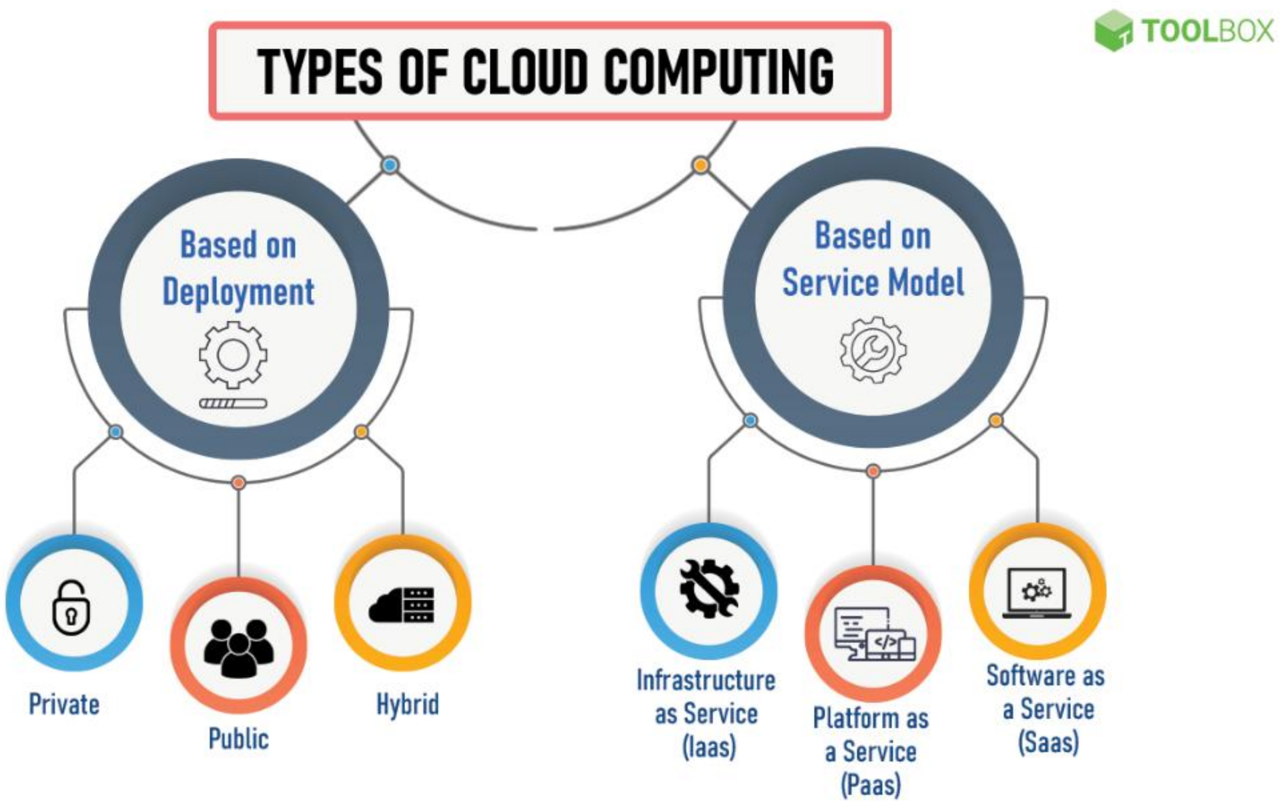
- Cloud Pricing Models
- Load Balancing
- Performance
- Security & Privacy



Cloud Computing Architecture

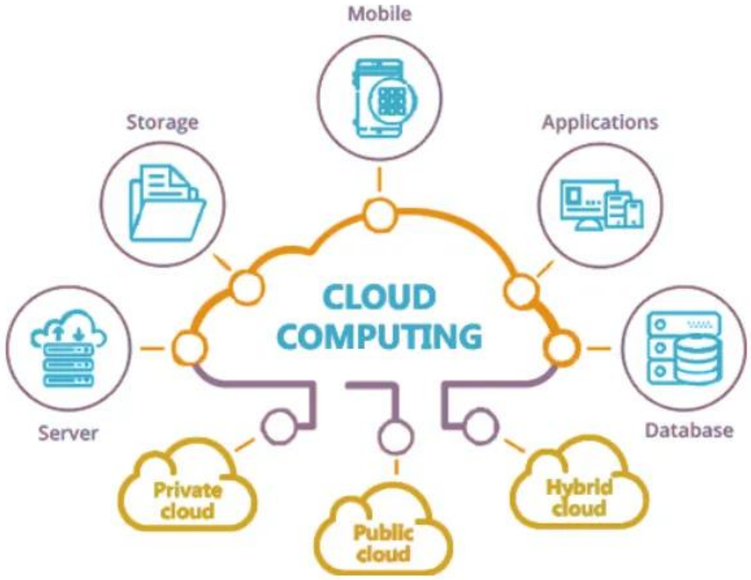


Types of Cloud Computing



Cloud Pricing Models

What Are Cloud Cost Models?



Common Cloud Pricing Models

- Pay-As-You-Go
- Spot Instances
- Reserved Instances
- Subscription-Based
- Volume Discounts
- Savings Plans
- Hybrid/Multi-Cloud

Common Cloud Pricing Models

Pricing Model	Description	Pros	Cons
Pay-As-You-Go	Pay only for what you use, with no long-term commitment	High flexibility; suitable for variable workloads	Higher unit costs; not cost-efficient for steady usage
Spot Instances	Bid on unused capacity at steep discounts	Extremely low prices; good for fault-tolerant workloads	No availability guarantee; risk of interruption
Reserved Instances	Commit to specific resources for 1–3 years for a discount	Significant savings for predictable workloads	Inflexible; risk of overprovisioning
Subscription-Based	Fixed recurring fee for a set of resources over a term	Predictable costs; bundled services may add value	Potential overpayment if usage drops; hard to scale mid-term
Volume Discounts	Per-unit prices drop as usage increases	Cost-effective for large-scale operations	Requires sustained growth; risk of underutilization
Savings Plans	Commit to a consistent spend in exchange for broad discounts	Flexible across services; good balance of savings and agility	Requires usage forecasting; overcommitment reduces value
Hybrid/Multi-Cloud	Combine public/private clouds and providers	Optimizes cost and performance by workload type	Complex cost management; integration and monitoring challenges

Which Is the Best Model For You? Key Considerations

- **Workload Predictability**

Workload Type	Recommended Pricing Models	Reasoning
Predictable	Reserved Instances, Subscription-Based	Long-term planning allows commitment in exchange for savings
Unpredictable	Pay-As-You-Go, Spot Instances	Allows flexible scaling with no commitment
Mixed (Both Types)	Hybrid Approach (Mix of Reserved + On-Demand/Spot)	Matches pricing models to workload volatility

Which Is the Best Model For You? Key Considerations

- **Budget Constraints**

Budget Type	Recommended Pricing Models	Reasoning
Tight/Fixed	Subscription-Based, Reserved Instances	Predictable billing reduces risk of cost overrun
Flexible/Variable	Pay-As-You-Go, Spot Instances	Enables adaptive spending and scaling

Which Is the Best Model For You? Key Considerations

- **Application Criticality**

Application Type	Recommended Pricing Models	Reasoning
Mission-Critical	Reserved Instances, Subscription-Based	Guarantees resource availability and minimizes risk
Non-Critical	Spot Instances, Pay-As-You-Go	Cost-effective even with potential for interruption

Which Is the Best Model For You? Key Considerations

- ## Scalability Requirements

Scalability Needs	Recommended Pricing Models	Reasoning
High/Variable Scaling	Pay-As-You-Go, Spot Instances	Flexible, real-time scaling without commitment
Low/Stable Requirements	Reserved Instances, Subscription-Based	Cheaper for consistent usage over time

Which Is the Best Model For You?
Key Considerations

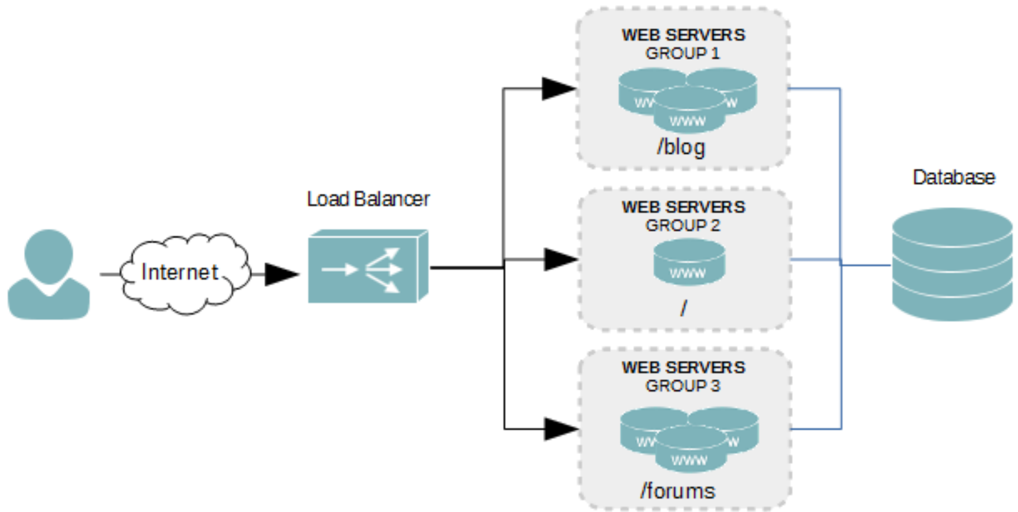
- **Long Term Planning**

Planning Horizon	Recommended Pricing Models	Reasoning
Clear Long-Term Roadmap	Reserved Instances, Subscription-Based	Maximizes savings and supports accurate budgeting
Uncertain/Short-Term	Pay-As-You-Go, Spot Instances	Retains flexibility; avoids lock-in

Cloud Cost Optimization Strategies

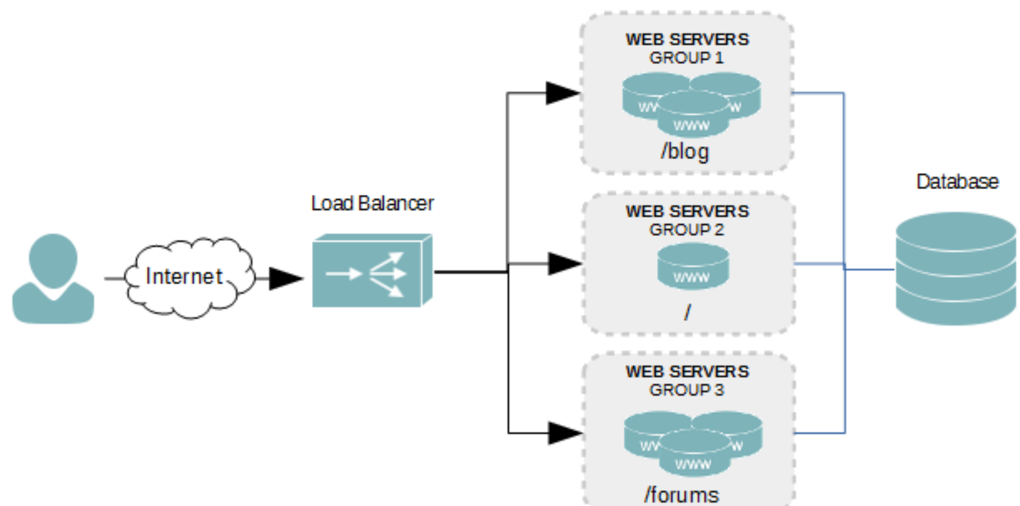
- Monitoring and Analysis
- Resource Rightsizing
- Utilizing Discounts
- Automated Scaling

Load Balancing in Cloud Computing

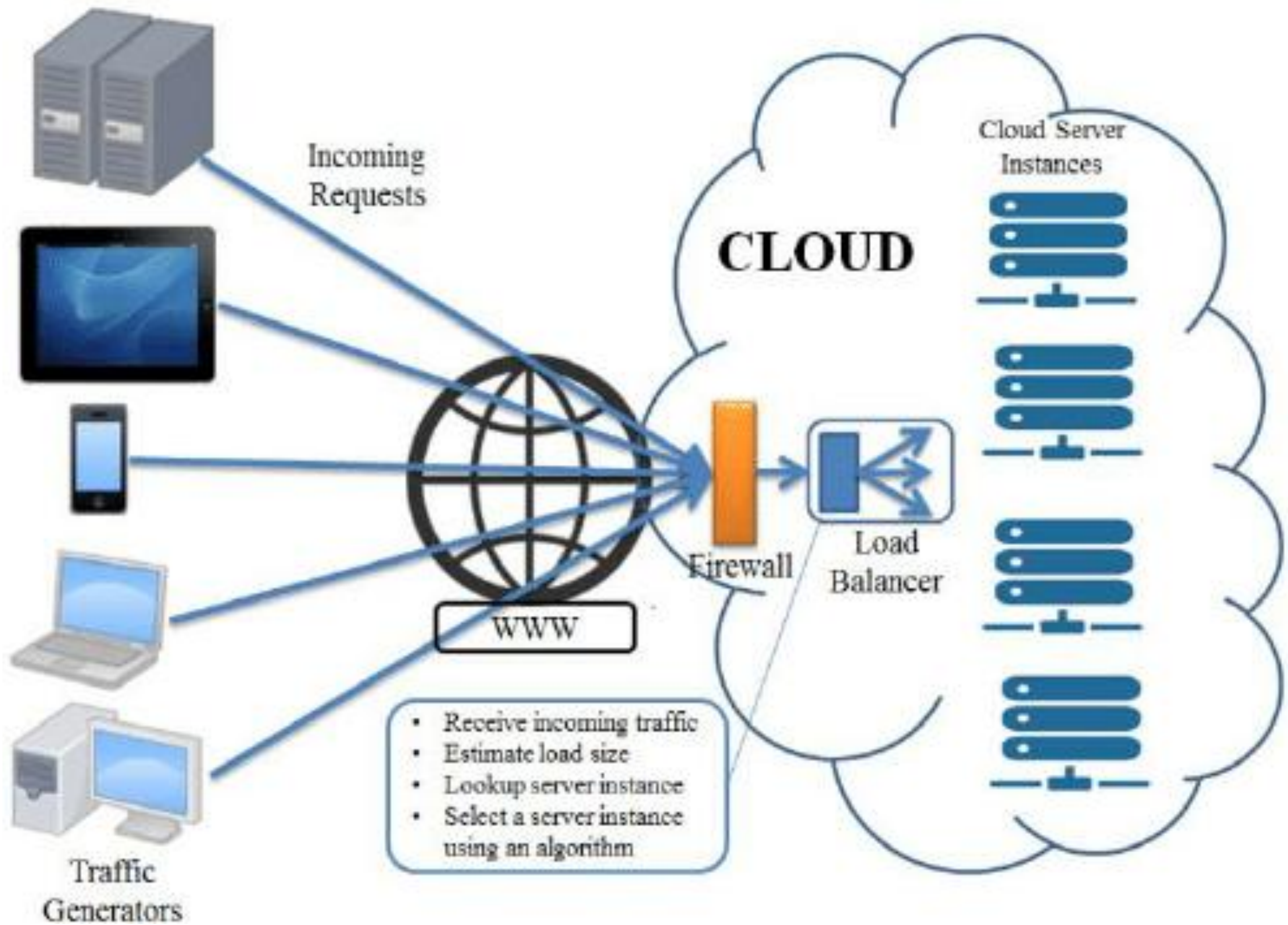


Load Balancing in Cloud Computing

- Load balancing is an essential technique used in cloud computing to optimize resource utilization and ensure that no single resource is overburdened with traffic
- It is a process of distributing workloads across multiple computing resources, such as servers, virtual machines, or containers, to achieve better
  - Performance
  - Availability
  - Scalability



Load Balancing in Cloud Computing



Load Balancing in Cloud Computing

- Load balancing can be implemented at various levels, including the network layer, application layer, and database layer
- The most common load balancing techniques used in cloud computing are Mainly
  - **Network Load Balancing:** This technique is used to balance the network traffic across multiple servers or instances. It is implemented at the network layer and ensures that the incoming traffic is distributed evenly across the available servers.
  - **Application Load Balancing:** This technique is used to balance the workload across multiple instances of an application. It is implemented at the application layer and ensures that each instance receives an equal share of the incoming requests.
  - **Database Load Balancing:** This technique is used to balance the workload across multiple database servers. It is implemented at the database layer and ensures that the incoming queries are distributed evenly across the available database servers.

Advantages of Load Balancing

1. **Improved Performance:** Load balancing helps to distribute the workload across multiple resources, which reduces the load on each resource and improves the overall performance of the system.
2. **High Availability:** Load balancing ensures that there is no single point of failure in the system, which provides high availability and fault tolerance to handle server failures.
3. **Scalability:** Load balancing makes it easier to scale resources up or down as needed, which helps to handle spikes in traffic or changes in demand.
4. **Efficient Resource Utilization:** Load balancing ensures that resources are used efficiently, which reduces wastage and helps to optimize costs.

Disadvantages of Load Balancing

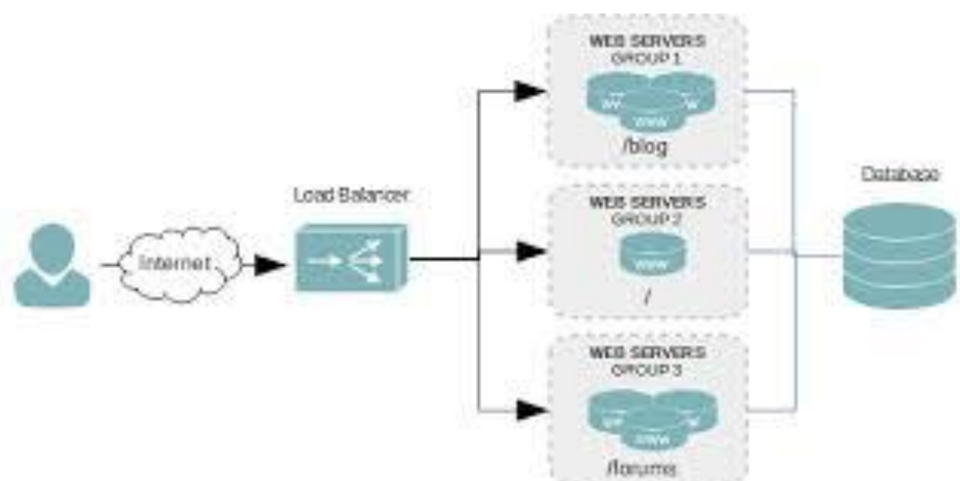
1. **Complexity**: Implementing load balancing in cloud computing can be complex, especially when dealing with large-scale systems. It requires careful planning and configuration to ensure that it works effectively.
2. **Cost:** Implementing load balancing can add to the overall cost of cloud computing, especially when using specialized hardware or software.
3. **Single Point of Failure:** While load balancing helps to reduce the risk of a single point of failure, it can also become a single point of failure if not implemented correctly.
4. **Security:** Load balancing can introduce security risks if not implemented correctly, such as allowing unauthorized access or exposing sensitive data.

Categories of Load Balancing Solutions

- **Software-based load balancers**
  - Software-based load balancers run on standard hardware (desktop, PCs) and standard operating systems.
- **Hardware-based load balancers**
  - Hardware-based load balancers are dedicated boxes which include Application Specific Integrated Circuits (ASICs) adapted for a particular use. ASICs allows high speed promoting of network traffic and are frequently used for transport-level load balancing because hardware-based load balancing is faster in comparison to software solution.

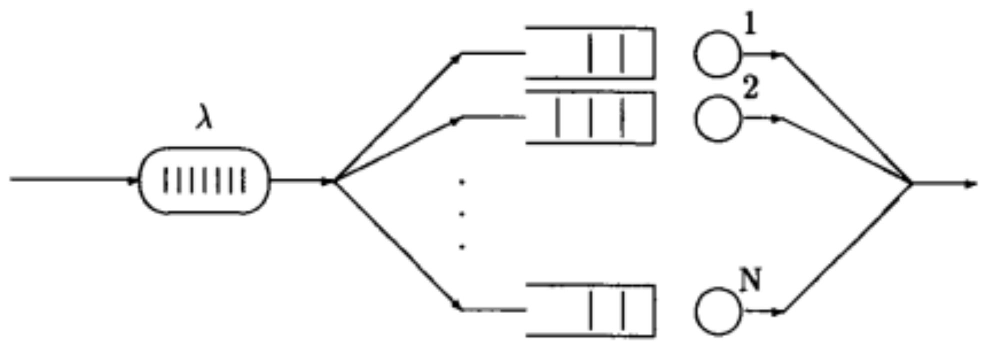
Smart Load Balancing

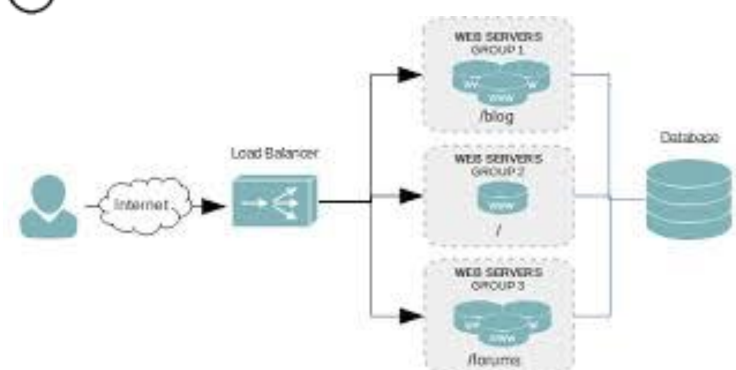
- Metrics
- A dispatcher does smart load balancing by utilizing
  - Server availability
  - Workload
  - Capability
  - Other user-defined criteria to regulate where to send a request



Load Balancing Algorithms

- Class Assignment
  - Propose a Load Balancing Algorithms by first defining appropriate Metrics/Parameters
  - Justify your solution





Research Challenges in Cloud Computing

- Architecture and Infrastructure Challenges
- Security, Privacy & Trust
- Resource Management & Optimization
- Intelligent Cloud Management
- Networking & Performance Challenges
- Cloud Sustainability and Green Computing
- …

Challenges in Cloud Computing

Security & Privacy in Cloud Computing

References

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